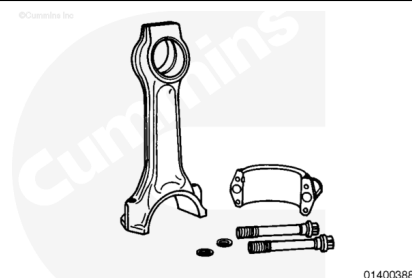


Trainings ▾

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

The number on the connecting rod must be the same as the number on the connecting rod cap. To reduce the possibility of engine damage, never assemble a new connecting rod cap to a old connecting rod or an old connecting rod cap to a new connecting rod.

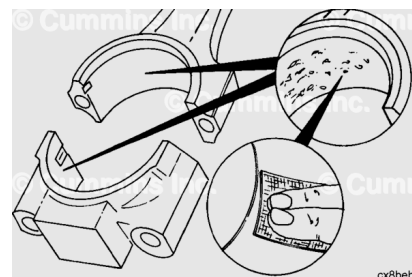
Remove the capscrews, washers, and the connecting rod cap from the connecting rod.

Clean the parts with solvent and dry with compressed air.

Inspect the connecting rod caps and capscrews for nicks, cracks, burrs, scratches, or fretting.

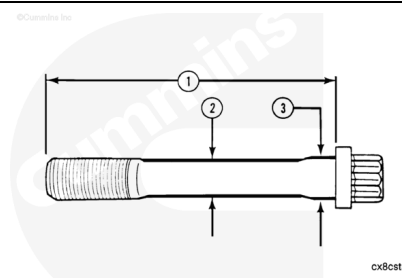
The connecting rod and connecting rod cap **must** be replaced as an assembly if any fretting damage is visible on either piece.

Inspect the bearing seating surface for nicks or burrs. If it is **not** possible to remove any nicks or burrs with a fine crocus cloth, the connecting rod **must** be replaced.



Check the threads of the connecting rod capscrews for damage. Check under the capscrew heads for cracks.

Measure the connecting rod capscrews.



Capscrew Length (1)		
mm		in
117.09	MIN	4.610
117.86	MAX	4.640

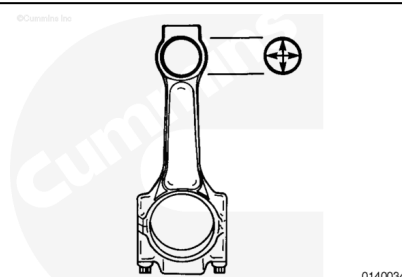
Capscrew Outside Diameter (2)		
mm		in
15.27	MIN	0.601
15.37	MAX	0.605

Capscrew Outside Diameter (3)		
mm		in
17.35	MIN	0.683
17.45	MAX	0.687

If the capscrews are **not** within specifications, the capscrews **must** be replaced.

Measure the connecting rod wrist pin inside diameter.

Connecting Rod Wrist Pin Bearing Inside Diameter		
mm		in
60.99	MIN	2.401
61.04	MAX	2.403



The bushing **must** be precision machined after installation. If machining capability is available, the bushing can be replaced. Reference the Alternate Repair Manual, Bulletin 3379035.

Calibrate

⚠ CAUTION ⚠

The connecting rod must be installed and tightened to specification or the measurement will be incorrect.

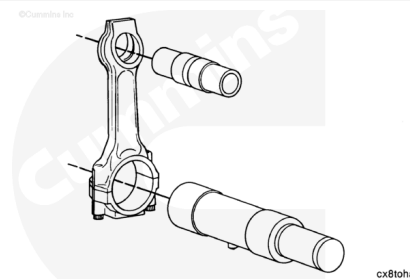
A connecting rod of known length, bend, and twist, called a master connecting rod, is required to calibrate the fixture. A new connecting rod with a known distance from center of the crank pin to the center of the piston pin end (connecting rod length) can also be used.

Install the appropriate mandrel in the piston pin end of the master connecting rod.

Note : Be sure the locating pin in the mandrel for the crank pin end is touching the connecting rod cap directly opposite the beam of the connecting rod.

Install the mandrel in the crank pin bore of the master connecting rod.

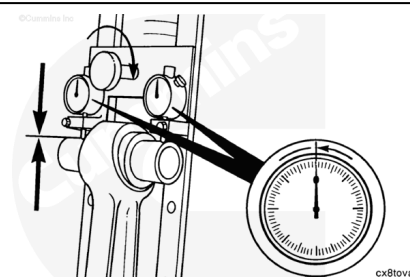
Tighten the mandrel so it is centered in the bore correctly.



Install the master connecting rod in the cranking fixture.

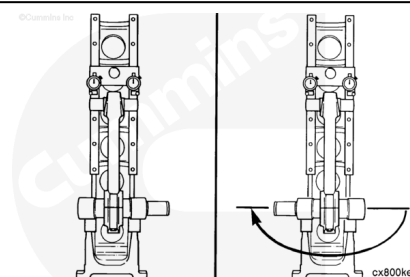
Loosen the knob and move the indicator bracket until both indicators touch the top of the piston pin end mandrel. Move the bracket toward the mandrel until the indicator needles have moved approximately 0.25 mm [0.010 in]. Turn the knob to tighten the bracket.

Adjust the indicator needles to "0". Move the **master** connecting rod in and out to confirm the "0" setting.



Remove the master connecting rod from the fixture.

Rotate the connecting rod 180 degrees horizontally and install it in the checking fixture.

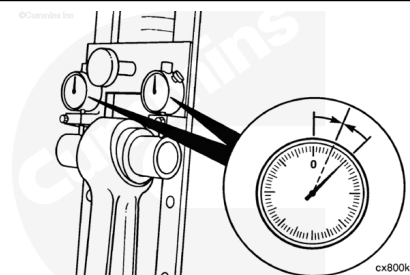


⚠ CAUTION ⚠

Do not adjust the indicators during this step of the procedure or the measurement will be incorrect.

Be sure the indicators are at "0" setting.

If the indicator needles do **not** return to the "0", adjust the indicator so that the "0" moves one half of the indicated difference from the needle to the "0" established during the previous step.



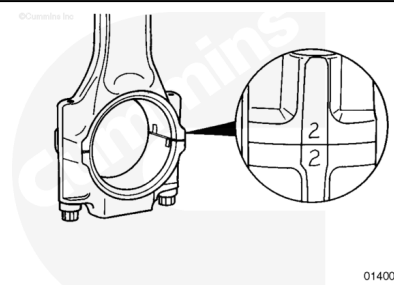
If the needle is more than 0.10 mm [0.004 in] from the "0" established during the previous step, check for dirt or burrs on the mandrels and fixture. If the "0" is still **not** within specification, check to be sure that the master connecting rod is **not** damaged.

After completing the calibration of the fixture, remove the mandrels from the master connecting rod.

Bend and Twist Inspect

⚠ CAUTION ⚠

Use a vise with brass jaws to hold the connecting rod. Notches, scratches, or dents in the I-beam area will cause engine failure.



01400395

⚠ CAUTION ⚠

The number on the connecting rod must be the same as the number on the connecting rod cap. To reduce the possibility of engine damage, never assemble a new connecting rod cap to a old connecting rod or an old connecting rod cap to a new connecting rod.

⚠ CAUTION ⚠

To reduce the possibility of engine damage, the connecting rod must be assembled with the capscrews tightened to specifications before stamping the cylinder identification number on the connecting rod. Always stamp a new connecting rod with the cylinder number of the connecting rod being replaced.

Lubricate the connecting rod capscrews with clean engine oil.

Assemble the connecting rod, connecting rod cap, washers, and capscrews.

Tighten the capscrews.

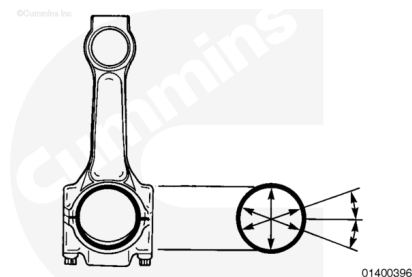
Refer to Procedure 001-054 in Section 1 (</qs3/pubsys2/xml/en/procedures/18/18-001-054-tr.html>) for proper torque values.

Use a dial bore indicator to measure the inside diameter within a 20 degree arc from each side of the parting line.

Measure the inside diameter at 90 degrees from the parting line.

Connecting Rod Bearing Bore Inside Diameter		
mm		in
107.993	MIN	4.2516
108.019	MAX	4.2528

If any of the three measurements are **not** within specifications, the connecting rod **must** be repaired or replaced.



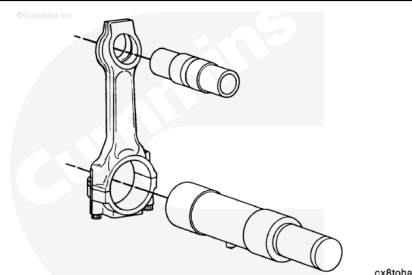
⚠ CAUTION ⚠

The connecting rod cap must be installed and tightened to specification or the measurements will not be correct.

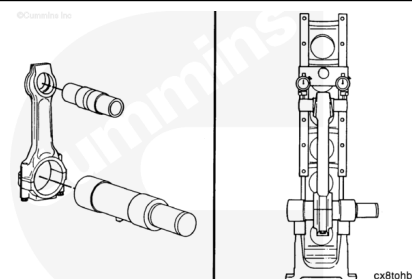
Install the appropriate mandrel in the piston pin end of the connecting rod.

Make sure the locating pin in the mandrel for the crank pin end is touching the connecting rod cap, directly opposite the beam of the connecting rod.

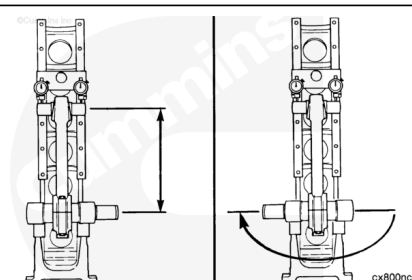
Install the mandrel in the crank pin bore of the connecting rod. Center the mandrel in the bore and tighten it.



Install the connecting rod into the fixture.



Record the indicator reading. The difference of the calibrated "0" of the indicators **must** be added or subtracted from the known length of the master connecting rod to determine the length of the connecting rod being measured.

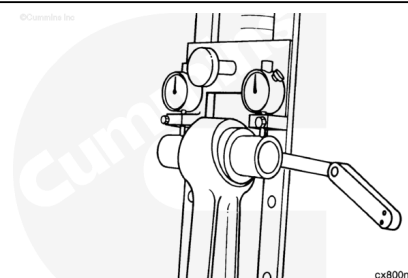


Connecting Rod Length		
mm		in
289.69	MIN	11.405
289.74	MAX	11.407

If the connecting rod length is **not** within specifications, the connecting rod **must** be replaced or the piston pin bushing replaced and machined.

⚠ CAUTION ⚠

Never attempt to straighten a connecting rod using heat or force. The rod will break eventually and cause extensive engine damage.



Record indicator readings.

Remove the connecting rod from the fixture. Turn the connecting rod 180 degrees horizontally.

Compare the indicator readings with those recorded in the previous step. The difference between the indicator readings is the amount of bend in the connecting rod.

Connecting Rod Bend		
mm		in
0.010	MIN	0.004
0.203	MAX	0.008

If the connecting rod bend is **not** within specification with the bushing installed, the bushing can be removed and bend measured again. If the connecting rod is **not** within specifications with the bushing removed, the connecting rod **must** be replaced.

Check the fixture and mandrel in the piston pin end for a gap between the two. If there is any twist in the connecting rod, the mandrel will **only** touch one side of the fixture.

Hold the end of the mandrel that is touching the fixture firmly against the fixture.

Use a feeler gauge to measure the gap between the mandrel and the fixture. The amount of gap between the mandrel and the fixture is the amount of connecting rod twist.

Maximum Connecting Rod Twist

	mm		in	
Bushing Installed	0.25	MIN	0.010	
Bushing Removed		0.51	MAX	0.020

If the connecting rod twist is **not** within specifications with the bushing installed, the bushing can be removed. If the connecting rod twist is **not** within specifications with the bushing removed, the connecting rod **must** be replaced.

Magnetic Crack Inspect

Use a magnetic particle testing machine.

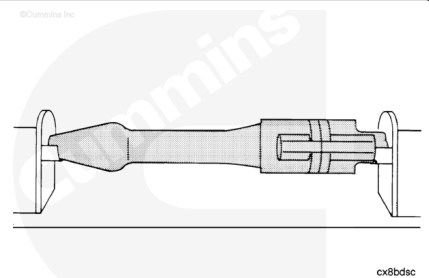
The connecting rod and connecting rod cap **must** be assembled during this check.

Use the residual method and apply head shot amperage.

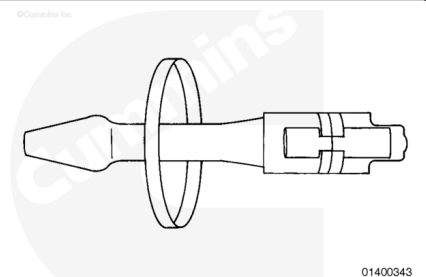
Adjust the amperage to 1500 ampere VDC or rectified VAC.

Check for cracks.

If the connecting rod is cracked, it **must** be replaced.



Use the residual method and apply coil shot amperage.



Amperage (Ampere Turns)	
Minimum	2600 VDC or Rectified VAC
Maximum	2600 VDC or Rectified VAC

An ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

Check for cracks.

If the connecting rod is cracked, it **must** be replaced.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.

⚠ CAUTION ⚠

The connecting rod must be demagnetized completely and cleaned thoroughly. Any small particles will cause engine damage.

Demagnetize the connecting rod.

Clean the connecting rod with solvent or steam.

